

Shubhojyoti Datta Chaudhuri

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EDUCATION

Master of Science in Computer Science (AI/ML Track)

University at Buffalo, The State University of New York, Buffalo, NY

Aug 2025 - Jun 2027

GPA: 3.5 / 4.0

Bachelor of Engineering in Computer Science

Chandigarh University, India

Aug 2021 - Jul 2025

GPA: 3.2 / 4.0

EXPERIENCE

Graduate Student Developer (CSE 611 Project) | Team of 4 | LookThePart

Jan 2026 – Present

- Designed a Django-based fashion discovery platform enabling users to shop outfits from TV shows and movies, designing 15+ relational database models including Entertainment, Characters, Products, and Posts.
- Developed 50+ RESTful API endpoints using Django REST Framework supporting authentication, product search, post sharing, and collection management for mobile application integration.
- Integrated AWS S3 cloud storage and automated media processing pipelines using the Pillow library, enabling scalable image storage with dynamic cropping and thumbnail generation.

Public Safety Aide | University at Buffalo Police Department

Dec 2025 – Present

- Monitored campus safety operations across a university community of 30,000+ students, assisting UB Police in non-emergency response coordination and maintaining compliance with established safety protocols.
- Logged and reported incident data using standardized documentation procedures, supporting patrol coordination and improving accuracy of campus safety records.
- Assisted officers during large campus events and routine patrols, helping maintain situational awareness and safe operations in high-traffic environments.

SKILLS

- Languages:** Python, SQL, JavaScript, C++, HTML, CSS
- Frameworks & Libraries:** Django, Django REST Framework, LangChain, PySpark, Hadoop, Flutter, OpenCV
- Machine Learning:** Regression, Classification, Clustering, XGBoost, Feature Engineering, Model Evaluation
- Databases & Tools:** MySQL, FAISS, Docker, Git, AWS S3, Figma
- Concepts:** REST APIs, Distributed Systems, Data Processing, RAG Pipelines, Vector Databases

PROJECT EXPERIENCE

Distributed Car Price Prediction System | Hadoop, PySpark

2025

- Configured a distributed Hadoop ecosystem using Docker Compose to simulate a 3-node cluster (NameNode + DataNodes) enabling scalable HDFS data processing.
- Applied PySpark feature engineering techniques including StringIndexer, OneHotEncoder, and VectorAssembler to transform high-cardinality categorical vehicle attributes into ML-ready vectors.
- Trained a Gradient Boosted Tree model improving prediction accuracy from 71% to 99% (R^2) while reducing mean absolute error to approximately \$520.

DocChat – RAG Based Document QA System | LangChain, Gemini, FAISS

- Built a Retrieval-Augmented Generation (RAG) system enabling natural language querying across complex PDF documents.
- Implemented FAISS vector indexing to enable sub-second semantic retrieval across embedded document chunks.
- Improved answer accuracy by designing hybrid search pipelines and structured prompt chains to reduce hallucinated responses.

Telemedicine Platform | Flutter

- Developed a cross-platform telemedicine application enabling remote doctor-patient consultations and appointment workflows.
- Designed appointment scheduling with real-time availability tracking and automated booking management.
- Implemented a secure API layer employing encryption protocols to safeguard patient data, reducing potential security vulnerabilities by 99%.

CERTIFICATIONS

- Elite + Silver Certification – Introduction to Internet of Things (NPTEL – IIT Kharagpur)
- Introduction to Computer Vision and Image Processing – IBM (Coursera)
- Data Visualization with Tableau – UC Davis (Coursera)
- Discrete Mathematics – NPTEL (IIT Ropar)